Deploying a Sovereign SatsPath Authority

This guide explains how to deploy `satspathd` as a **sovereign single-domain authority** or a **multi-tenant hosted provider**.

SatsPath allows `name@yourdomain.com` to be realistically deployable without hiding the underlying security model.

Onboarding Workflow

1. Initialize Operator and DNS Records

Run the CLI operator tool to generate your operator keys and output the required DNS records:

```
satspath server init yourdomain.com
```

Add the printed `TXT` records to your DNS provider (e.g., Cloudflare, Route53, Bind).

2. Verify DNS and Trust Quorum

Once DNS records propagate, verify the DNSSEC chain, operator key, and transparency log:

```
satspath server check yourdomain.com
```

3. Deploy with Docker Compose

We provide a reproducible deployment via Docker Compose. Private user keys **never** reside on the server and are generated locally by users.

Edit `docker-compose.yml` to set `SATSPATH_AUTHORITY_DOMAIN` and deploy:

```
docker-compose up -d
```

4. Reverse Proxy TLS Example (Caddy)

We strongly recommend placing `satspathd` behind a TLS-terminating reverse proxy.

Caddyfile example:

```
yourdomain.com {
    reverse_proxy localhost:4848
}
```

Backup & Restore

- **Backup**: Backup the `satspath-data` docker volume. Backups exclude user identity private keys because those keys never reside on the server.
- **Restore**: Restoring a backup preserves the transparency `log_id`, pins, histories, and witness continuity.

Upgrade Guidance

Upgrading `satspathd` is as simple as pulling the latest container image and restarting:

```
docker-compose pull
docker-compose up -d
```